Confidence-Weighted Pair Trading via DCC-eGARCH

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Pair trading

- A pair trade holds offsetting positions in two assets whose prices share a long-run equilibrium, and profits when a temporary deviation from that equilibrium closes.
- Three design choices govern such a strategy, namely which pairs to trade, when to enter and exit, and how much capital each pair receives.
- We hold the first fixed and test the second and third against each other.
- We estimate time-varying hedge ratios with the Kalman filter of Kalman (1960).
- We model spread volatility by a generalized autoregressive conditional heteroskedasticity (GARCH) model in the exponential form (eGARCH) of Nelson (1991).
- We embed it in the dynamic conditional correlation (DCC) framework of Engle (2002).

Research question

Question

1. Does optimizing the entry and exit thresholds alone improve pair trading?
2. Does optimizing them together with a confidence weighting of the pairs improve it?

Finding

1. Optimizing the thresholds alone lowers the Sharpe ratio from 1.32 to 1.17.
2. Together with the confidence weighting, optimized thresholds deliver the highest total return, the highest Sharpe ratio at 1.43 and the lowest drawdown.

We backtest cointegration-based pair trading on a pool of US equities and exchange-traded funds (ETFs) over an evaluation window from August 2021 to December 2023.

Data

GDX	GDXJ	GLD	AAPL	GOOGL	AMD
NVDA	CSCO	ORCL	TTWO	EA	HYG
LQD	JNK	SLV	USLV	SIVR	USO
QQQ	SPY	VOO	VDE	VTI	VDC
KXI	IBB	VHT	VNQ	IYR	MSFT
PG	TMF	UPRO	WFC	JPM	GS
CVX	XOM	INTC	COST	WMT	T
VZ	CMCSA	AMZN			

Asset pool screened for pairs formation.

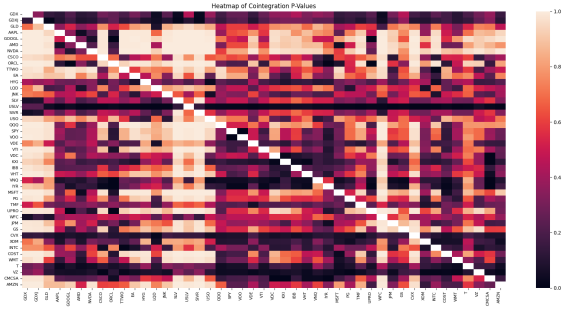
- The pool is 45 US equities and ETFs of large market capitalization and high liquidity.
- Daily closing prices run from January 2012 to December 2023, for names with a complete price history.
- An 80-20 split gives a training set from January 2012 to 5 August 2021 and a testing set from 6 August 2021 to December 2023.

Cointegration

$$s_t = y_t - (\beta_t x_t + \alpha_t)$$

- Two price series x_t and y_t are cointegrated when the spread s_t is (weakly) stationary, with β_t the hedge ratio and α_t the intercept.
- We retain a pair when the two-step test of Engle and Granger (1987) rejects the null of no cointegration at the 5% level.

Engle-Granger p-values



- 73 of the $\binom{45}{2}$ candidates are significant and carried forward.
- The 73 pairs span 38 distinct tickers.

Engle-Granger p-values for every candidate pair on the training sample.

Darker cells indicate stronger evidence of cointegration.

Time-varying hedge ratio & Half-life

$$\phi_t = \phi_{t-1} + \omega_t, \qquad \omega_t \sim N(0, \delta I),$$
$$y_t = (1, x_t)\,\phi_t + \nu_t, \qquad \nu_t \sim N(0, \sigma_\nu^2),$$

- A hedge ratio estimated once by least squares will drift out of date, so we treat $\phi_t = (\alpha_t, \beta_t)^\top$ as a latent state following a random walk.
- We estimate it with the Kalman filter, with state noise scale $\delta = 10^{-3}$, I the identity matrix and observation variance $\sigma_\nu^2 = 2$.

$$\Delta s_t = \kappa\, s_{t-1} + \epsilon_t, \qquad H = -\frac{\ln 2}{\kappa}$$

- Regressing the spread increment on its lagged level gives the half-life H, the number of days over which a deviation is expected to halve.
- Each pair uses its own H, rounded to whole days, as its rolling window length.

Volatility clustering

- No training spread is consistent with normality at the 5% level under either the Shapiro and Wilk (1965) or Jarque and Bera (1980) test; Quantile-quantile plots show heavier tails than the Gaussian (Appendix A.1).
- We therefore adopt a Student t conditional distribution throughout.
- The eGARCH extends the autoregressive conditional heteroskedasticity (ARCH) model of Engle (1982).
- It admits an asymmetric response to shocks, the leverage effect, and models the logarithm of the variance, so that positivity holds for any parameter values.

Exponential GARCH

$$\varepsilon_t = h_t \eta_t, \qquad \eta_t \sim \text{Student-t}(0,1)$$

$$\ln h_t^2 = \omega_h + \sum_{i=1}^{p} \left(a_i \left| \eta_{t-i} - \mathbb{E}|\eta_{t-i}| \right| + \lambda_i \eta_{t-i} \right) + \sum_{j=1}^{q} b_j \ln h_{t-j}^2$$

- h_t is the conditional volatility, ω_h the constant, a_i the ARCH coefficients, b_j the GARCH coefficients and λ_i the leverage coefficients.
- Over $p, q \in \{1, 2, 3, 4, 5\}$, the Bayesian information criterion is minimized at $p = 3$ and $q = 1$ on its mean and median (Appendix B.1).

Dynamic conditional correlation

$$\mathbf{v}_t = D_t^{-1}(\mathbf{s}_t - \boldsymbol{\mu}), \qquad Q_t = (1 - \theta_1 - \theta_2)\bar{Q} + \theta_1 \mathbf{v}_{t-1}\mathbf{v}_{t-1}^\top + \theta_2 Q_{t-1}$$

- The 73 spreads are not independent, so we estimate their volatilities jointly.
- $\mathbf{v}_t$ standardizes the spreads $\mathbf{s}_t$ at time t, with $\boldsymbol{\mu}$ their conditional means and D_t the diagonal matrix of conditional volatilities.
- Q_t is the quasi-correlation matrix, θ_1 and θ_2 govern its persistence and $\bar{Q}$ is its unconditional level.
- Fitted on the training spreads with multivariate Student t innovations, the persistence parameters are $\hat{\theta}_1 = 0.123$ and $\hat{\theta}_2 = 0.672$.
- Over the testing sample, the fitted model is iterated forward to extract each spread's conditional volatility for the trading signal at every date.

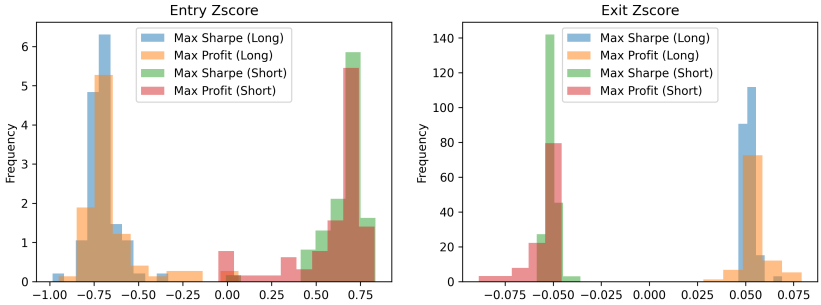
Trading signal

$$z_{i,t} = \frac{s_{i,t} - \bar{s}_{i,t}}{h_{i,t}}$$

long entry	$z_{i,t} < \mathrm{LET}_i < z_{i,t-1}$,	long exit	$z_{i,t-1} < \mathrm{LETX}_i < z_{i,t}$,
short entry	$z_{i,t-1} < \mathrm{SET}_i < z_{i,t}$,	short exit	$z_{i,t} < \mathrm{SETX}_i < z_{i,t-1}$.

- Spread i is standardized by its rolling mean $\bar{s}_{i,t}$ over the half-life window and its conditional volatility $h_{i,t}$.
- A position opens when $z_{i,t}$ crosses an entry threshold and closes when it crosses the corresponding exit threshold.
- LET, LETX, SET and SETX are the long entry, long exit, short entry and short exit thresholds, and crossings are strict.
- We optimize the four thresholds separately for each pair by the Nelder-Mead simplex, maximizing either profit or the Sharpe ratio.

Optimized entry and exit thresholds



Distribution of the optimized entry and exit thresholds.

- The long entry concentrates between -0.70 and -0.65 and the long exit near 0.05.
- The short entry concentrates between 0.60 and 0.65 and the short exit near -0.05.
- Summary statistics under both objectives are in Appendix B.2.

Confidence score

$$\gamma_t(i) = h_i \cdot \mathrm{Coint}_t(i, r), \qquad \gamma_p(i) = \frac{h_i}{\mathrm{Coint}_p(i, r)}$$

- Equal weighting ignores that the 73 pairs differ in how reliably they revert, so we propose a confidence score.
- The strength of cointegration over a trailing window of $r = 250$ trading days is measured by the absolute test statistic $\mathrm{Coint}_t(i, r)$ or the p-value $\mathrm{Coint}_p(i, r)$.
- h_i is the forecast volatility of spread i. When a position is opened at an extreme z-score, a more volatile spread is more likely to return toward its mean than to move further away.
- Each weight is capped at 4x the equally weighted share, $c = \frac{4}{n}$ (about 5.5% at $n = 73$).
- Weights are assigned through passes in which the effective cap tightens as capital is committed (Appendix C.1).

Strategies

- EW-FE Equal weights and the same fixed thresholds for every pair, entering at ± 0.8 and exiting at ∓ 0.05.
- EW-OE Equal weights and the thresholds optimized for profit.
- OW-OE Confidence weights γ_p and the same thresholds.

- All three run through the same BackTrader engine.
- Orders are submitted at market and fill at the opening price of the following bar. Transaction cost is 0.1% of traded notional on each side, and no slippage is applied.
- The evaluation window covers 593 trading days, 23 August 2021 to 29 December 2023.

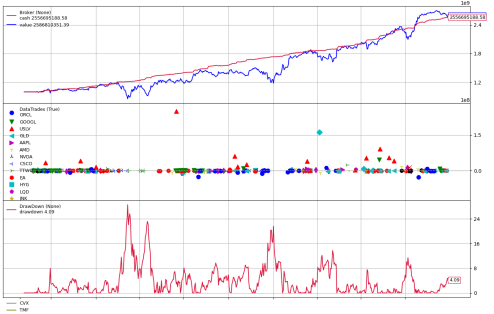
Strategy metrics

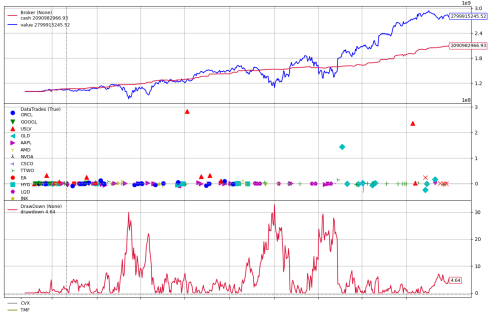
	EW-FE	EW-OE	OW-OE
Total return	158.68%	179.99%	202.22%
Sharpe ratio	1.32	1.17	1.43
Maximum drawdown	28.58%	32.75%	25.59%
System quality number	3.68	2.72	3.28
Number of trades	540	386	393

Strategy metrics over the evaluation window, net of transaction costs.

- Optimizing the thresholds does not pay for itself.
- Despite a higher return, EW-OE has a Sharpe ratio of 1.17, lower than 1.32 for EW-FE.
- With the confidence weighting, OW-OE earns the highest return, the highest Sharpe ratio at 1.43 and the lowest drawdown.

Backtest paths for EW-FE and EW-OE



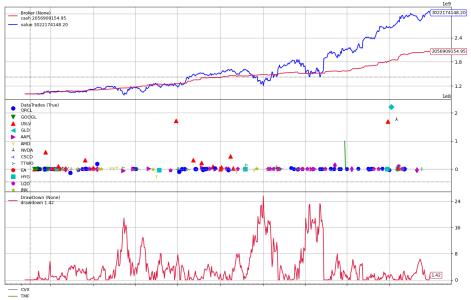


Backtest paths for EW-FE, left, and EW-OE, right.

Each panel reports portfolio value, cash, per-trade profit and loss and drawdown.

- Optimizing the thresholds improves returns.
- It raises the annualized volatility from 33.7% to 44.2%.
- It raises the maximum drawdown from 28.58% to 32.75%.

Backtest paths for OW-OE



Backtest paths for OW-OE.

- OW-OE earns the highest return, while its volatility of 36.2% and maximum drawdown of 25.59% are much lower than those of EW-OE.
- Focusing on the pairs with strong cointegration and volatile spreads improves both risk and return.

Conclusion

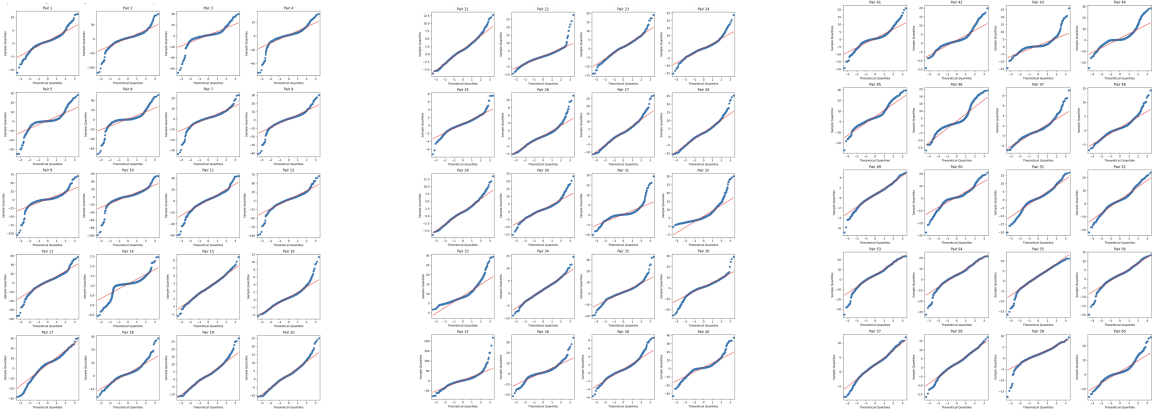
- We backtest cointegration-based pair trading with DCC-eGARCH spread volatility on 73 pairs of US equities and ETFs from August 2021 to December 2023.
- We optimize the entry and exit thresholds and weight pairs by a confidence score based on the rolling cointegration statistic and forecast volatility.
- Optimizing the thresholds pair by pair improves returns but increases volatility and drawdown by more, so the Sharpe ratio falls from 1.32 to 1.17.
- Combined with the confidence score, threshold optimization delivers the highest total return, the highest Sharpe ratio at 1.43 and the lowest drawdown.

References

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Code and replication data are available at *GitHub*.

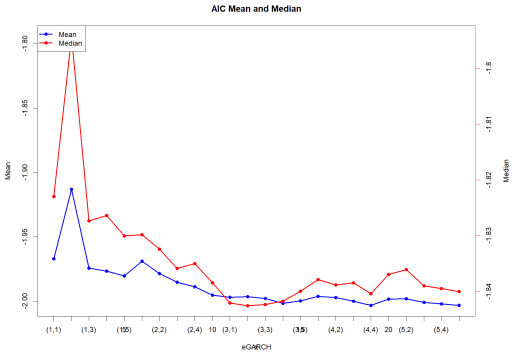
Appendix A.1. Quantile-quantile plots

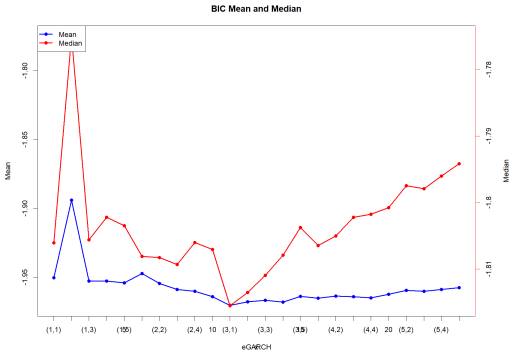


Quantile-quantile plots of training spreads against the Gaussian.

The plots show heavier tails than the Gaussian.

Appendix B.1. Information criterion by eGARCH order





Akaike and Bayesian information criterion by eGARCH order.

- The Bayesian criterion is minimized at $p = 3$ and $q = 1$ on both mean and median.
- The Akaike criterion draws a similar conclusion.

Appendix B.2. Summary statistics of the optimized thresholds

	MaxProfit	long_entryZscore	long_exitZscore	short_entryZscore	short_exitZscore
mean	3.437518	-0.647286	0.054060	0.587717	-0.054488
std	1.636079	0.177435	0.007117	0.220871	0.008078
min	1.212882	-0.949097	0.028142	-0.049572	-0.088670
25%	2.414950	-0.729579	0.050584	0.548860	-0.054809
50%	2.897327	-0.693437	0.051872	0.680232	-0.051939
75%	3.530215	-0.640698	0.056442	0.708750	-0.050185
max	8.043060	0.064196	0.079131	0.829552	-0.045588

	MaxSharpe	long_entryZscore	long_exitZscore	short_entryZscore	short_exitZscore
mean	2.033816	-0.700840	0.051767	0.659499	-0.050989
std	0.501662	0.087618	0.003221	0.116720	0.003282
min	0.452877	-0.983603	0.046328	-0.007080	-0.058483
25%	1.810431	-0.746912	0.049935	0.620172	-0.052893
50%	2.021001	-0.706552	0.051105	0.694531	-0.050934
75%	2.359007	-0.665103	0.053075	0.724888	-0.049531
max	3.235803	-0.332021	0.069006	0.834995	-0.035816

Summary statistics of the thresholds optimized for maximum profit (left) and maximum Sharpe ratio (right).

Appendix C.1. Weight calculation

$$w_i = R_k\, c \qquad \text{for every } i \notin A_k \text{ with } \frac{\gamma_i}{\sum_{j \notin A_k} \gamma_j} > R_k\, c, \text{ subject to } \sum_{i=1}^{n} w_i = 1$$

- Pass $k+1$ assigns these weights, with $c = \frac{4}{n}$ the cap at four times the equally weighted share and γ_i the confidence score of pair i.
- A_k is the set of pairs assigned a weight after k passes, starting with $A_0 = \emptyset$, and $R_k = 1 - \sum_{i \in A_k} w_i$ the remaining unassigned weight.
- The passes repeat until no further pair qualifies. Since R_k falls at every pass, the effective cap tightens as capital is committed.
- Pairs never assigned receive the smallest assigned weight, and the vector is rescaled to satisfy the constraint.